

JAMES HARDY

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Professional Summary

Software design engineer and IT professional with over two decades of experience spanning kernel and device-driver debugging, full-stack application development (C/C++, C#, Java), large-scale data analytics (Cosmos), enterprise technical support, and cross-platform (Windows/Linux/Android) engineering. Proven track record of promotion within Microsoft's engineering organization, entrepreneurial business ownership, and hands-on technical leadership.

Skills

- Kernel & device driver debugging for Windows (Windbg, Kd); experience across ARM, x86, and AMD64
- C / C++ application and OS-level programming for Windows and Linux; familiar with the STD library
- C# programming for the .NET Framework; Java application programming (including phone/PDA and Android development)
- SQL/ODBC database programming in C++, C#, and Java (MS SQL, MySQL)
- Highly proficient with TCP/IP; strong LAN/WAN design, setup, and troubleshooting; network analysis and packet-sniffing tools
- IT operations: router/switch configuration, computer hardware & software configuration and troubleshooting in a corporate environment
- Large-scale data mining and analytics tooling with Cosmos (streams, xFlow jobs, insight reports)
- Strong technical writing skills: knowledge base articles, training content, and course material development

Certifications & Permits

- Microsoft Certified Systems Engineer (MCSE) — Windows 2003, Windows 2000, and NT 4.0
- C Programming Certificate, University of Washington Continuing Education Program

Work Experience

DataAnnotations *currently employed*

AI model training and rating for Data Annotations. Work includes evaluating multiple models and their output against themselves and against other AI agents.

- Evaluate and train models through turn based model prompt and response.
- Author system prompts and input prompts for models and rate their responses for accuracy and effectiveness.
- Author model grading guidelines for AI training to allow AI to rate and evaluate another models output on tasks.

Microsoft Corporation — Software Design Engineer II (Application Compatibility Team) *Jul 2014 – Dec 2022*

Multifaceted role on the Application Compatibility team: authored design change requests for managerial review of OS code changes, tested applications and drivers for post-upgrade functionality, and performed data-mining work with Cosmos, including authoring streams, xFlow jobs, and insight reports.

- Built a Dynamic API Sampling engine (kernel and service components) enabling logging of all Windows function calls for code coverage analysis; redesigned the API sample packaging library, added full Unicode character-set support, and built a stress test suite to verify engine operation.
- Built a Windows Error Reporting analysis tool from the ground up using statistical word analysis to categorize user feedback and parse uploaded error reports for relevant details such as error codes; designed the desktop interface, a SQL database for intermediate results, and a background processor pulling and parsing results from Cosmos to SQL.
- Authored XML code for the migration library to remove or alter applications with post-upgrade compatibility issues.
- Coded a migration engine plug-in to remove registry entries for MSI packages.
- Contributed to a migration engine plug-in that simplified application removal by programmatically invoking application uninstallers.
- Wrote Cosmos scope queries and authored insight reports.

Microsoft Corporation — SDET II, Device Compatibility Team*Jul 2012 – Jul 2014*

Primary duties included kernel device driver debugging, writing kernel driver shims, and creating tools to automate device testing.

- Promoted from SDET to SDET II.
- Created and drove implementation of a framework using SQL to index driver API dependencies, focusing testing on devices affected by public API changes.
- Worked with third-party vendors to resolve issues with applications and device drivers; authored kernel driver shims to mitigate third-party code issues.
- Used kernel- and user-mode debugging tools (Windbg, Kd) to triage device driver failures in Windows 8 lab tests.
- Debugged across ARM, x86, and AMD64 architectures.
- Mentored new team members on debugging technique and kernel architecture.
- Tracked, prioritized, and triaged issues using Product Studio and WTT Test Studio.

Microsoft Corporation — Software Design Engineer in Test III (Prithvi Catalytic)*May 2011 – Jul 2012*

SDET with the Application/Device Compatibility team for Windows 8; kernel device driver debugging, third-party application compatibility testing and debugging, and issue prioritization.

- Promoted from SDET II to SDET III within 3 months of job commencement.
- Worked with third-party vendors to resolve application and device driver issues.
- Used kernel- and user-mode debugging tools (Windbg, Kd) to triage device driver failures in Windows 8 lab tests.
- Tracked, prioritized, and triaged issues using Product Studio and WTT Test Studio.
- Authored a team onboarding document with training materials, how-tos, and links to key resources.

Taxi / Click-A-Cab — Software Design Vendor*Jun 2009 – Oct 2010*

Vendor for a taxicab dispatch and fleet-management program commissioned by a Kitsap County cab company, integrated with Click-A-Cab, a mobile app enabling voiceless taxicab requests via a Java-based web service.

- Designed and implemented a MySQL-based database for customer and fleet record tracking.
- Designed and coded Taxi, a C# Windows application interfacing with MySQL via the .NET Framework, supporting call tracking, Google Maps integration, and lease/payment tracking and reporting.
- Designed, coded, and implemented Click-A-Cab's Java-based web servlets and JSPs.
- Designed and coded an Android mobile app enabling web-service-based taxicab requests.
- Built and maintained a companion website for the application.

El Chupacabra Restaurant — Owner*2005 – 2008*

Designed, built, owned, and operated a bar and restaurant with an entire build-out (electrical, plumbing, interior construction, appliance installation); reached approximately \$750,000 in annual gross revenue.

- Supervised, scheduled, hired, and terminated all front- and back-of-house staff (approx. 20 full- and part-time employees).
- Managed full accounting and payroll using QuickBooks and Excel; handled all ordering and stocking.
- Designed, implemented, and maintained a computerized POS system with 4 remote stations, integrated with Virtual PC for quick restore and remote network printing.
- Implemented a multi-subnet network for customer Wi-Fi access while protecting internal systems, plus VPNs for remote troubleshooting, system restore, and secure off-site bookkeeper/accountant access.
- Remodeled the restaurant interior, including a concrete countertop bar and custom woodwork; managed marketing and local ad placement.

World Ticketing Systems — Lead Programmer*Oct 2004 – Mar 2006*

Lead programmer for a multi-tiered event-ticketing platform; clients included Holland Ticketing, the Dutch National Soccer Association, and Billetix (Istanbul, Turkey). Met with customers to scope design/feature requests and translated them into deliverable specs and timelines for the Windows client and Linux back end.

- Developed design change requests in C/C++ and MFC C++ to customer specifications on the Win32 client.
- Added COM functionality to the Win32 client for interoperability with third-party programs and tools.
- Implemented MS SQL and ODBC code to enable third-party Point-of-Sale integration.
- Coded required changes in C on the Linux back end to support client-side design changes.

- Analyzed and fixed bugs across both Windows client and Linux server components.

Microsoft Corporation — CPR Engineer, File Systems and Networking *Jun 2000 – Oct 2004*

Worked with enterprise-level customers to determine root cause and deliver resolutions for networking and file-system issues in Windows client and server OSs; handled end-to-end troubleshooting and authored hotfixes. Received a Microsoft Ship It award for significant contributions to Windows XP.

- Analyzed and debugged operating system source code to find bugs in released operating systems; partnered with OS developers on fixes for service packs and hotfix releases.
- Developed test case programs to expose hard-to-reproduce issues.
- Analyzed memory dump files and live systems, locally and remotely, using various debugging tools to determine root cause.
- Used network analysis tools to troubleshoot customer issues and determine root cause.
- Wrote hotfix update patches (primarily C/C++) for operating systems and provided interim solutions ahead of service pack releases.
- Created and delivered training content for internal support personnel, external partners, and MVPs.
- Authored Knowledge Base articles detailing issue symptoms, causes, and step-by-step resolutions.

Microsoft Corporation — Senior Escalation Engineer *Apr 1998 – Jun 2000*

Troubleshooting of operating system server/client issues for Premier/enterprise-level customers and resolved technical issues beyond the scope of primary support personnel; focus areas included OS setup, mass deployment, and file-system issues.

- Provided technical support and solutions to premier customers, establishing root cause across Microsoft and third-party software.
- Mentored internal and external primary support personnel.
- Developed and delivered training content and triage guidance for primary support personnel.
- Set up computer hardware and software lab environments to reproduce and pinpoint customer issues.

Keane Inc. — Technical Support Engineer *1997 – 1998*

Provided front-line technical support in a call-center environment; served as vendor representative with the Windows 98 Beta Team, triaging beta tester issues and feedback for the Windows development team.

- Provided phone- and web-based technical support to Microsoft customers.
- Designed and coded an Emergency Boot Disk creation tool included on the Windows 98 CD.
- Triaged beta tester reports to identify OS bugs; worked with lab test engineers to reproduce issues.
- Mentored co-workers in real-time customer issue resolution.

Greyhound Lines Inc. — Mechanic *Prior to 1997*

- Performed repairs on all components of Greyhound buses, including engines, air conditioning, and electrical systems.